Sports Tournament Ranking System

Python Programming – Activity-Based Learning (ABL) Assignment

1. Problem Statement

Store sports match results and team scores, calculate total points and tie-break values, and produce rankings according to a specified competition policy.

2. Objectives

• Store each match result. • Calculate matches played, wins, draws, losses, goals for, goals against, goal difference, and points. • Sort teams using the competition tie-break rules.

3. Assumptions and Competition Policy

A win earns 3 points, a draw earns 1 point for each team, and a loss earns 0 points. Teams are ranked by (1) total points, (2) goal difference (GF – GA), (3) goals scored (GF), and (4) team name alphabetically. The sample tournament contains six matches and four teams.

4. Algorithm

```
1. Create an empty dictionary to store each team's statistics.
2. Read each match and create team records if they do not exist.
3. Update played matches, goals for, and goals against.
4. Award points and update win/draw/loss counts based on the score.
5. Sort teams by points descending, goal difference descending,
   goals scored descending, and team name ascending.
6. Display the ranked table and a completion message.
```

5. Flowchart (Text Representation)

```
START
  |
  v
Read match results
  |
  v
Update team statistics and points
  |
  v
All matches processed? -- No --> Read next match
  |
 Yes
  v
Calculate goal difference
  |
  v
Sort using competition tie-break policy
  |
  v
Display ranking table
  |
  v
END
```

6. Python Source Code

```
# Sports Tournament Ranking System
# Policy: Win = 3 points, Draw = 1 point, Loss = 0 points.
# Tie-breakers: total points, goal difference, goals scored, then team name.

matches = [
    ("Falcons", "Tigers", 2, 1),
    ("Eagles", "Sharks", 1, 1),
    ("Falcons", "Eagles", 0, 2),
    ("Tigers", "Sharks", 3, 1),
    ("Falcons", "Sharks", 2, 0),
    ("Tigers", "Eagles", 1, 1),
]

POINTS_WIN = 3
POINTS_DRAW = 1
```

```
POINTS_LOSS = 0

teams = {}

def get_team(name):
    if name not in teams:
        teams[name] = {
            "played": 0, "won": 0, "drawn": 0, "lost": 0,
            "gf": 0, "ga": 0, "points": 0
        }
    return teams[name]

# Process each match and update both teams.
for home, away, home_goals, away_goals in matches:
    home_stats = get_team(home)
    away_stats = get_team(away)

    home_stats["played"] += 1
    away_stats["played"] += 1
    home_stats["gf"] += home_goals
    home_stats["ga"] += away_goals
    away_stats["gf"] += away_goals
    away_stats["ga"] += home_goals

    if home_goals > away_goals:
        home_stats["won"] += 1
        away_stats["lost"] += 1
        home_stats["points"] += POINTS_WIN
        away_stats["points"] += POINTS_LOSS
    elif home_goals < away_goals:
        away_stats["won"] += 1
        home_stats["lost"] += 1
        away_stats["points"] += POINTS_WIN
        home_stats["points"] += POINTS_LOSS
    else:
        home_stats["drawn"] += 1
        away_stats["drawn"] += 1
        home_stats["points"] += POINTS_DRAW
        away_stats["points"] += POINTS_DRAW

# Competition policy: points > goal difference > goals scored > team name.
ranking = sorted(
    teams.items(),
    key=lambda item: (
        -item[1]["points"],
        -(item[1]["gf"] - item[1]["ga"]),
        -item[1]["gf"],
        item[0]
    )
)

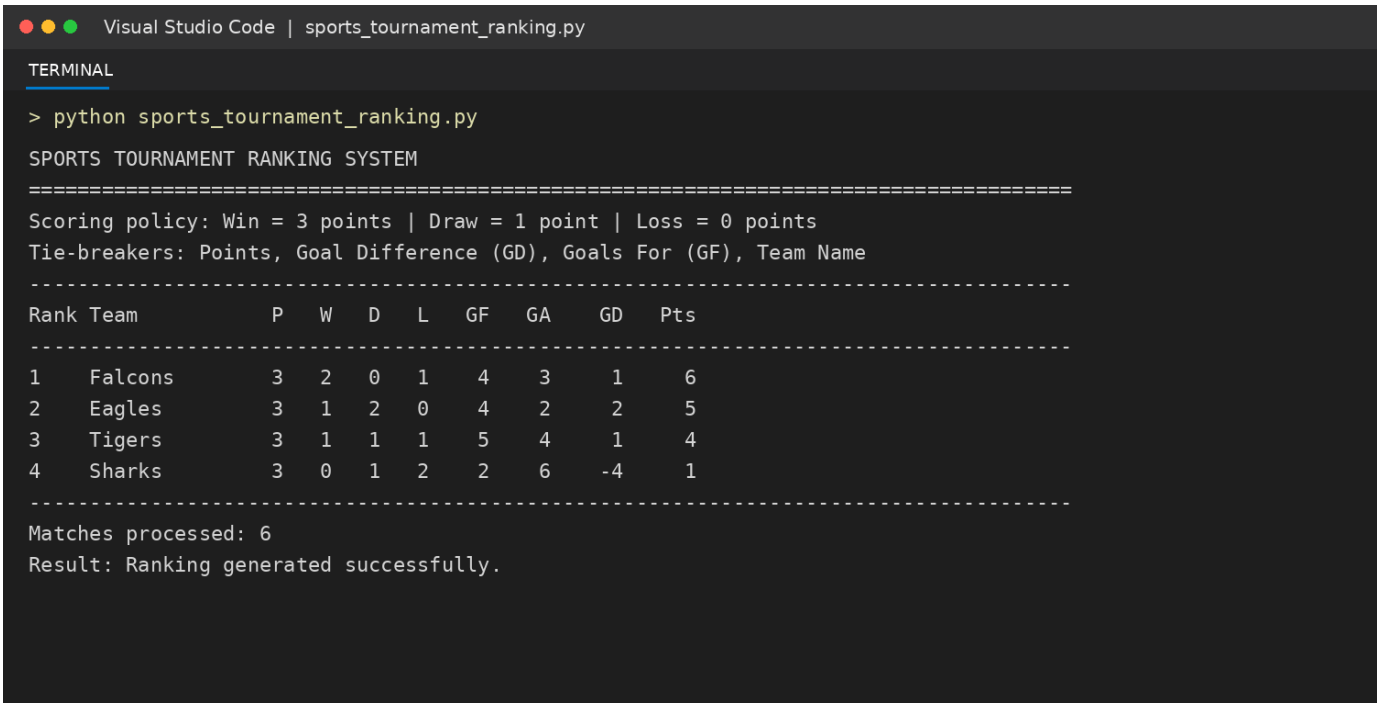
print("SPORTS TOURNAMENT RANKING SYSTEM")
print("=" * 86)
print("Scoring policy: Win = 3 points | Draw = 1 point | Loss = 0 points")
print("Tie-breakers: Points, Goal Difference (GD), Goals For (GF), Team Name")
print("-" * 86)
print(f"{'Rank':<5}{'Team':<12}{'P':>4}{'W':>4}{'D':>4}{'L':>4}{'GF':>5}{'GA':>5}{'GD':>6}{'Pts':>6}")
print("-" * 86)

for rank, (name, stats) in enumerate(ranking, start=1):
    gd = stats["gf"] - stats["ga"]
    print(f"{rank:<5}{name:<12}{stats['played']:>4}{stats['won']:>4}"
          f"{stats['drawn']:>4}{stats['lost']:>4}{stats['gf']:>5}"
          f"{stats['ga']:>5}{gd:>6}{stats['points']:>6}")

print("-" * 86)
print(f"Matches processed: {len(matches)}")
print("Result: Ranking generated successfully.")
```

7. Execution Output Screenshot

The following screenshot-style capture shows the program output in a Visual Studio Code terminal layout. Run the included Python file in VS Code to reproduce the output.



8. Test Cases

```
Sample match results:
Falcons 2–1 Tigers; Eagles 1–1 Sharks; Falcons 0–2 Eagles;
Tigers 3–1 Sharks; Falcons 2–0 Sharks; Tigers 1–1 Eagles.

Expected checks:
• Every team has played 3 matches.
• A win adds 3 points, a draw adds 1 point, and a loss adds 0.
• Goal difference is calculated as goals for minus goals against.
• Sorting follows points, goal difference, goals scored, then team name.
```

9. Conclusion

The program successfully processes match results, calculates team statistics and points, applies the specified tie-break rules, and prints a final tournament ranking. The scoring policy can be changed by updating the constants and sorting key.